

Nada Essam Mohammed Amin

AI Engineer | Data Analyst | Business Intelligence Analyst | Digital Marketing & Business Strategy Enthusiast

ahes64033@gmail.com | +201033689664 | Alexandria, Egypt | [GitHub](#)

Summary

AI Engineer specializing in Artificial Intelligence, Machine Learning, Data Analytics, and Business Intelligence, with hands-on experience in developing AI-powered systems, intelligent automation, and scalable software solutions. Strong research aptitude combined with analytical thinking and creative problem-solving to transform complex challenges into innovative, data-driven solutions. Passionate about advancing AI through research, education, and real-world applications while contributing to both academic excellence and industry innovation.

Education

Faculty of Artificial Intelligence , Kafr El-Sheikh University	Oct. 2022 – Jun. 2026
Bachelor of Artificial Intelligence	Graduation project: A+ CGPA: 3.41 / 4.00
Relevant Coursework: Machine Learning • Deep Learning • Computer Vision • NLP • Data Analytics • Business Intelligence • Database Systems • Software Engineering	

Professional Experience

AI & Data Science Instructor | Oct 2025 – Dec 2025

- Delivered hands-on AI, ML, Data Analytics, and BI training through project-based learning.
- Mentored students in developing AI solutions using Python and modern AI tools.

AI & Machine Learning Trainee | Oct 2025 – Dec 2025

- Built end-to-end ML pipelines using Python for predictive analytics and model optimization.
- Applied supervised and unsupervised learning to solve real-world data problems.

Business Intelligence Analyst Trainee – ITI | Aug 2025 – Sep 2025

- Built interactive Power BI dashboards and analyzed business data using SQL and Excel.
- Delivered data-driven insights through visualization and KPI reporting.

Flutter Developer Intern – Konecta | Jan 2025 – Apr 2025

- Developed scalable Flutter applications using clean architecture and responsive UI design.
- Integrated Firebase, SQL, authentication, and REST APIs into mobile applications.

Mobile Application Development Trainee – DEPI | Apr 2024 – Oct 2024

- Developed cross-platform Flutter applications using clean architecture and state management.
- Implemented Firebase, REST APIs, local storage, and Agile development practices.

AI Trainee – Huawei HCIA-AI Learning | Aug. – Sep. 2024

- Completed hands-on AI, ML, and Deep Learning training using the Huawei HCIA-AI curriculum.
- Applied AI fundamentals to real-world use cases while strengthening problem-solving and analytical skills.

Research Program Participant – New York Academy of Sciences | Sep. – Nov. 2021

- Conducted scientific research through a global research program, strengthening critical thinking and analytical skills.
- Collaborated in an international research environment while developing scientific communication and teamwork skills.

PROJECTS

NassaQ — AI Document Management System | PaddleOCR • LangChain • RAG • NLP • Streamlit Jan. – Jun. 2026

- Developed an enterprise AI platform integrating 6+ intelligent modules, including bilingual OCR, semantic search, Retrieval-Augmented Generation (RAG), document classification, RBAC, and analytics dashboards.
- Supported 2 languages (Arabic & English), 4+ document formats, and implemented storage optimization to improve document retrieval, security, and team productivity.

ChronosInsight — AI Manuscript Restoration | Python • Vision Transformers (ViTs) • GANs • OpenCV • NLP Sep. 2025

- Developed a deep learning pipeline using ViTs, GANs, OCR, and NLP for manuscript restoration and handwriting recognition.
- Implemented semantic indexing and document classification using unsupervised learning.

EcoPulse — Carbon Intelligence Platform | Python • Machine Learning • Time-Series Forecasting • Power BI • SQL Oct. 2025

- Built an AI-powered sustainability platform leveraging IoT, geospatial, and time-series data for emissions forecasting.
- Developed Power BI dashboards and ML forecasting models for sustainability insights.

Whisker Bites — Food Ordering App | Flutter • Dart • Firebase • SQL • REST APIs

Dec. 2024

- Developed a scalable food ordering application with authentication, dynamic menus, cart, and secure checkout.
- Integrated SQL database architecture for efficient management of users, orders, menus, and payments.

Plantiful — Smart Plant Care App | Flutter • Dart • Firebase • Weather API • Location Services

Oct. 2024

- Developed a Flutter-based smart plant care app integrating Weather API, geolocation, smart irrigation, and plant monitoring.
- Implemented intelligent plant care recommendations to optimize irrigation efficiency and enhance sustainable gardening.

Electro Plants — Bioelectric Energy Research | MFCs • Bioenergy • Clean Energy • Research • Green Tech

Oct. 2022

- Investigated Microbial Fuel Cells (MFCs) for renewable bioelectric energy generation.
- Evaluated bioenergy generation techniques to improve renewable energy efficiency for sustainable agricultural applications.

Awards, Research & Competitions

Ideal Student Competition | Dec. 2025

- Recognized for academic excellence, leadership, and community engagement at the university level.

UGRG – Nile University Research Program | Aug. 2024

- Conducted sustainability research on cactus-based bioelectric energy generation for desert rehabilitation.

Climate Science Olympiad | Jun. 2023

- Designed an atmospheric water harvesting solution using silica gel to support sustainable desert irrigation.

Smart Innovative Competition | Feb. 2023

- Developed a thermoelectric energy harvesting solution for sustainable off-grid power generation.

APPX – Zewail City | Dec. 2021

- Built an Arduino-based smart irrigation system using soil moisture sensors for water optimization.

Diamond Challenge for High School Entrepreneurs | Oct. 2021

- Designed an innovative oxygen extraction concept from seawater to address COVID-19 medical supply challenges.

TECHNICAL SKILLS

- **Programming Languages:** Python • SQL • Dart • Java • C++ • Kotlin
- **Artificial Intelligence:** Machine Learning • Computer Vision • NLP • Generative AI • RAG • OCR • Prompt Engineering
- **Data Analytics & BI:** Power BI • Pandas • NumPy • Data Analytics • Data Visualization • KPI Reporting • Time-Series Forecasting
- **Mobile Development:** Flutter • Firebase • REST APIs • Clean Architecture • State Management
- **Marketing & Business:** Market Research • Marketing Strategy • Content Marketing • Social Media Marketing • Customer Analysis
- **Tools & Platforms:** Git • GitHub • Jupyter Notebook • Google Colab • Android Studio • Figma
- **Soft Skills:** Leadership • Problem Solving • Critical Thinking • Communication • Teamwork • Creativity

LEADERSHIP & ACTIVITIES

- **HR Member** | IYNA Gharbia • TEDx KFS
- **PR Member** | STEM Hotspot • KFS Astronomy Club